

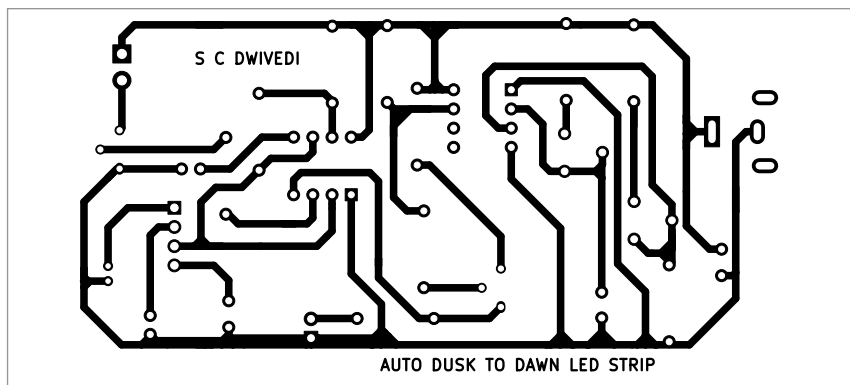


Fig. 3: Actual-size PCB

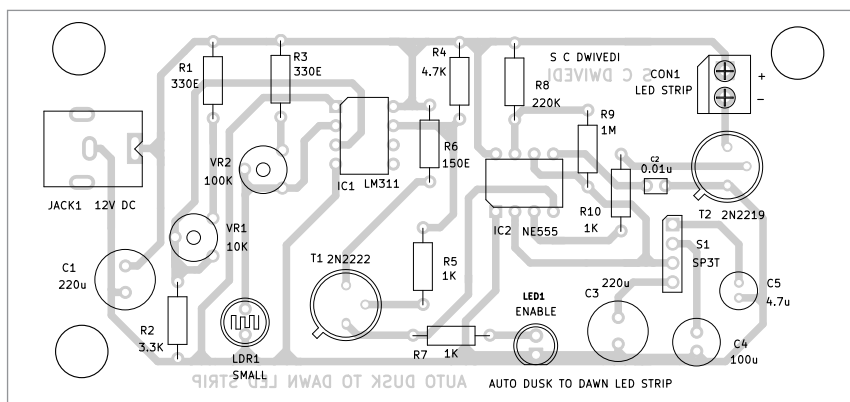


Fig. 4: Components layout

noise and ripple from the supply. The LDR, together with VR1, VR2, R2, and R3, forms a voltage divider that senses ambient light intensity. Preset VR2 allows adjustment of the light-sensitivity level (the threshold between day and night).

The sensed voltage is fed to pin 2 of IC1 (LM311 comparator). When the ambient light falls below the preset level (set at pin 3 of IC1), the comparator output changes state. Resistor R4 acts as a pull-up resistor for the open-collector output of the LM311, while R5 limits the base current to transistor T1. When transistor T1 conducts, the NE555 timer IC2 receives a voltage at pin 4 (reset pin) and becomes enabled. LED1 serves as an enable indication for IC2. Capacitor C2 stabilises the control voltage pin of the 555 timer against noise.

The time delay is determined by resistors R8, R9, switch S1 (SP3T), and capacitors C3, C4, or C5. Using

switch S1, different timing capacitors can be selected to obtain short, medium, or long delay periods. The output of the NE555 at pin 3 drives transistor T2 (2N2219) through base resistor R10. This transistor acts as a power switch for the 12V LED strip connected to CON1.

The circuit's operation is simple. During daylight, the LDR's resistance is low, causing the voltage at pin 3 (set by VR2) to remain below the reference voltage at pin 2 (set by VR1). As a result, the comparator output stays low, transistor T1 remains cut off, and the NE555 timer is not enabled. Consequently, transistor T2 also remains cut off, and the LED strip stays switched off.

At night, the LDR's resistance increases, causing the voltage at pin 3 to rise above the reference voltage at pin 2. The LM311 output goes high, turning transistor T1 on and enabling the NE555 timer. The enable LED (LED1) glows,

PARTS LIST	
<i>Semiconductors:</i>	
IC1	- LM311 voltage comparator
IC2	- NE555 timer IC
T1	- 2N2222 NPN transistor
T2	- 2N2219 NPN transistor
LED1	- 5mm LED
<i>Resistors (all 1/4-watt, $\pm 5\%$ carbon):</i>	
R1, R3	- 330 Ω
R2	- 3.3k Ω
R4	- 4.7k Ω
R5, R7, R10	- 1k Ω
R6	- 150 Ω
R8	- 220k Ω
R9	- 1M Ω
VR1	- 10k Ω Preset
VR2	- 100k Ω Preset
<i>Capacitors:</i>	
C1, C3	- 220 μ F, 25V electrolytic
C2	- 0.01 μ F ceramic
C4	- 100 μ F, 25V electrolytic
C5	- 4.7 μ F, 25V electrolytic
<i>Miscellaneous:</i>	
LDR1	- Light-dependent resistor (small type)
S1	- SP3T single-pole three-terminal switch
CON1	- 2-pin connector
JACK1	- 12V DC power jack
	- 12V DC regulated power supply/adaptor
	- PCB/General-purpose board
	- IC bases for LM311 and NE555 (recommended)
	- Connecting wires, solder, and enclosure
	- 12V LED strip, 60-LED/5M roll

indicating that the timer is active. The NE555 output goes high/low for a preset duration determined by the selected timing capacitor (C3, C4, or C5) and timing resistors (R8 and R9). During this period, transistor T2 conducts, supplying power to the LED strip, which turns on. After the set delay, the NE555 output goes low, transistor T2 cuts off, and the LED strip turns off. This on/off cycle repeats continuously throughout the night.

The output delay is selected using switch S1, which connects one of the timing capacitors into the circuit. For example, selecting C3 provides a long delay of about 295 seconds (approximately 4.9 minutes), suitable for extended lighting. Choosing C4 gives a medium delay of around 134 seconds (about 2.2 minutes), while selecting C5 results in a short delay of about 6.3 seconds for brief